

H.P. Marshall, Boise State University
Current / Pending
April 18, 2023

Current

NSF Major Research Instrumentation (MRI)

Long-range Airborne Snow and Sea Ice Observing System (LASSIOS)

Total: \$161,883 to BSU

Labor: CoPI Marshall (1 month), Research Tech (3 months), Student Funding (0)

9/15/2018-8/31/2024

NASA Instrument Incubator Program (IIP)

CMOS Ku-band Radar for Snow

Total: \$1,123,384 to BSU

Labor: PI Marshall (1 month), Research Tech (11 months), Student Funding (0)

2/1/2020-4/28/2024

USFS Rocky Mountain Research Station

Snow Temperature Modeling

Total: \$40,000

Labor: PI Marshall (0), Student Funding (14 months)

10/1/2021-9/30/2024

NSF Arctic Observing Network (AON)

NSFGEO-NERC: Collaborative Research: The Integrated Characterization of Energy, Clouds, Atmospheric state, and Precipitation at Summit (ICECAPS): Melt along Lagrangian Transects

Total: \$204,068 to BSU

Labor: CoPI Marshall (0.5 months), Research Tech (6 months), Student Funding (0)

1/1/2022-11/30/2024

NASA Terrestrial Hydrology Program (THP)

Evaluation of L-band InSAR using L-band GPR

Total: \$60,813

Labor: CoPI Marshall (0.5 months), Student Funding (0)

1/1/2022-12/31/2024

NASA Instrument Incubator Program (IIP)

Snow Water equivalent Wide swath SAR and InSAR (SNOWWI)

Total: \$354,698

Labor: CoPI Marshall (1 month), Student Funding (24 months)

1/1/2022-2/28/2025

NASA Terrestrial Hydrology Program (THP)

SnowEx 2022-23 Leadership Support

Total: \$137,968

Labor: PI Marshall (1 month), Student Funding (0)

6/1/2022-9/31/2024

NASA EPSCoR

CryoIdaho: Building Idaho's Cryosphere Research Community through Analysis of Terrain Effects on Snow and Ice Meltwater Fluxes

Total: \$750,000

Labor: CoPI Marshall (0.25 months), Student Funding (0)

1/1/2021-12/31/2023

NASA JPL

Support of SoOp Grand Mesa Field Campaigns

Total: \$60,366

Labor: PI Marshall (0.5 months), Student Funding (0)

1/1/2023-12/31/2023

Future Investigators in NASA Earth and Space Science and Technology (FINESST)

Evaluating NISAR's InSAR Potential in Snow Covered Regions, Student: Zach Keskinen

Total: \$148,643

Labor: PI Marshall (0 months), Student Funding (12 months)

9/1/2023-8/31/2026

DoD U.S. Army Cold Regions Research and Engineering Lab (CRREL)

Climate and natural hazards, snow-covered and mountain environment sensing research

Total: \$1,500,000

Labor: PI Marshall (2.25 months), Student funding (36 months)

9/1/2023-8/31/2025

Pending

NASA Earth Venture Suborbital – 4 (EVS-4)

SnowBridge: Improving seasonal snow mass and albedo in land surface models:

Leveraging a time series of coincident multi-sensor high-resolution airborne remote sensing campaigns and intensive field experiments

Total: No budget requested – if awarded, PI will manage ROSES call for Science Team.

Total budget for mission: \$15-30m

Labor: PI Marshall (6 months)

NASA Earth System Explorer (ESE)

SnowGlobe: NASA's Snow Constellation

Total: \$675,000 to Boise State

Total budget for mission: \$350m

Labor: Co-PI Marshall (1-2 months)

DoD U.S. Army Cold Regions Research and Engineering Lab (CRREL)

Snow strength mapping for mobility applications: combining field observations, radar remote sensing, and modeling to create local to global snow trafficability products

Total: \$3,284,206

Labor: PI Marshall (3 months), Student funding (36 months)

Bureau of Reclamation

Development of a Snow Water Equivalent "Landing Strip" for Use Towards Improved CBRFC Water Supply Forecasts

Total: \$459,723

Labor: PI Marshall (0.25 months)

[This Proposal] NASA High Priority Open Source Science

The SnowEx DB open-source project -- Standardized data access to maximize mission data use and accelerate research

Total: \$90,954

Labor: PI Marshall (0.25 months)